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The Hong Kong University of Science and Technology
Clear Water Bay, Kowloon
Hong Kong

Cindy Agriningsih Haruna

Education

MPhil in Civil and Environmental Engineering (Currently)
(Expected Sep 2028)
The Hong Kong University of Science and Technology

M.Sc. in Bio-Chemical Engineering and Technology
(July 2026)
Thammasat University, Thailand
GPA: 3.75/4.00

B.Sc. in Chemistry
(Feb 2023)
Halu Oleo University, Indonesia
GPA: 3.67/4.00

Skills & Expertise

- Analytical Chemistry
- Water Sampling and Analysis
- Biodegradable and Antibacterial Food Packaging Materials
- Agricultural and Biowaste Valorization
- FTIR, TGA/DTG, Mechanical Testing, and Antibacterial Analysis

Curriculum Vitae

Cindy Agriningsih Haruna

Graduate student in Civil and Environmental Engineering specializing in waste valorization, mass transfer mechanisms, and green nanocomposite development. Proven track record in material synthesis, characterization, and antibacterial testing for sustainable food packaging. Passionate about driving circular economy solutions by synthesizing cellulose-based biodegradable composites reinforced with food waste fibers to enhance the mechanical performance and environmental sustainability of molded packaging systems.

Professional Experience

Sep 2026–Current
Graduate Research Assistant
Department of Civil and Environmental Engineering, HKUST
- Research focus: Development of Cellulose-Based Biopolymer Composites Reinforced with Food Waste Fibers for Sustainable Take-away Packaging Applications
- PI: Prof. Samir Kumar Khanal

Aug 2024–June 2026
Graduate Research Assistant
Department of Bio-Chemical Engineering and Technology, SIIT, Thammasat University, Thailand
- Research focus: Synthesis of Biodegradable Composite Films from Bio-waste for Food Packaging Applications
- PI: Prof. Sandhya Babel

Sep 2019–Nov 2022
Undergraduate Research
Department of Chemistry, Halu Oleo University, Indonesia
- Research focus: Green synthesis of copper (Cu) nanoparticles using dragon fruit (Hylocereus polyrhizus) extract for antibacterial disinfectant
- PI: Dr. La Ode Ahmad Nur Ramadhan

Research Areas

- Biodegradable polymers and composite films
- Waste-to-value materials for food packaging
- Sustainable packaging technology
- Antibacterial activity of natural extracts

Curriculum Vitae

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Publications

1. **Haruna, C. A.,** & Babel, S. (2026). Circular Valorization of Durian Rind to Microcrystalline Cellulose and Shrimp Shells to Chitosan for Synthesis of Composite Antibacterial Biodegradable Packaging Films, Environmental Technology and Innovation, 43, 105053. doi.org/10.1016/j.eti.2026.105053
2. **Haruna, C. A.,** Malik, W. A., Rijal, M. Y. S., Watoni, A. H., & Ramadhan, L. O. A. N. (2022). Green Synthesis of Copper Nanoparticles Using Red Dragon Fruit (Hylocereus polyrhizus) Extract and Its Antibacterial Activity for Liquid Disinfectant. Jurnal Kimia Sains dan Aplikasi, 25 (10), 352–361. doi.org/10.14710/jksa.25.10.352-361
3. Nurhidayah, N., Ramadhan, L. O. A. N., Watoni, A. H., Kadir, L. A., Rahmatullah, M. D., & **Haruna, C. A.** (2024). Bacterial Cellulose Incorporated Zinc Phosphate Nanocomposite for Antibacterial Agent and Air Particulate Matter Filtration. Malaysian Journal of Science, 43 (1), 22–33. doi.org/10.22452/mjs.vol43no1.3
4. Ramadhan, L. O. A. N., Fanihu, S. L. C., Watoni, A. H., Rijal, R., Santoso, N. S. C., Rafiqa, W. O. N., **Haruna, C. A.,** & Arifin, Z. S. (2024). Green preparation of Cu nanoparticles with bioreductor of red dragon fruit peel extract: visible spectroscopy properties. Journal of Tropical Pharmacy and Chemistry, 8 (1) 93–100. doi.org/10.25026/jtpc.v8i1.633
5. Nurhidayah, N., Watoni, A. H., Kadir, L. A., Rahmatullah, M. D., **Haruna, C. A**., & Ramadhan, L. O. A. N. (2024). Synthesis of Zinc Phosphate with Ammonium Dihydrogen Phosphate Precursor and Dragon Fruit Peel Extract as a Sensitizer for Antibacterial Materials. AIP Conference Proceedings, 3068, 080008. The 4th International Conference on Chemistry and Material Sciences (IC2MS). doi.org/10.1063/5.0201959

Awards and Honors

1. **PostGraduate Studentship**–The Hong Kong University of Science and Technology–Sept 2026–Current
2. **TISR Scholarship (Thammasat International Student Requirement)**–Thammasat University–Thailand–Aug 2024–July 2026
3. **Outstanding Scholarship Program**–Ministry of Education–Indonesia–2020–2023
4. **Research Grant** (PKM-RE), Ministry of Research, Technology, and Higher Education–Indonesia–2021–2022
5. **Entrepreneurship Grant** (PKM-K), Ministry of Research, Technology, and Higher Education–Indonesia–2022

Conference Presentation (* denotes presenter)

1. **Haruna, C. A.***, & Babel, S. Development of Biodegradable Films from Durian Rind and Shrimp Shells for Sustainable Packaging Application. Sept 4, 2025, 7th EnvironmentAsia International Conference, Bangkok, Thailand.

Relevant Courses

Graduate Courses
(M.Sc in Bio-Chemical Engineering and Technology)

- Environmental Chemical Engineering

Undergraduate Main Courses
(B.Sc in Chemistry)

- Organic Chemistry
- Inorganic Chemistry
- Chemistry Physics
- Biochemistry
- Methods of Chemical Separation
- Synthesis of Organic Chemistry
- Synthesis of Inorganic Chemistry
- Environmental Chemistry
- Environmental Microbiology